

REACTION – 10 µL

Pipette top to bottom. **Enzymes last.**

- 6** ddH₂O (white rack)
- 1** 10× T4 DNA ligase buffer (red)
- 1** each DNA fragment (2 of them, 2 µL total)
- 0.5** T4 DNA ligase
- 0.5** BsaI

10 µL total

10 µL is deliberate – it is exactly what the transformation takes. Carry the whole reaction forward; do not split it.

THEN

1. Mix well and quick spin.
2. Label the tube with your assigned code.
3. Put every reaction from your section in **one thermocycler block**.
4. Run **main/GG1**.

PROGRAM MAIN/GG1

repeat 25×:

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37 °C – 2 min
16 °C – 5 min
45 °C – 10 min
80 °C – 10 min
16 °C – hold
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37 °C is BsaI cutting, 16 °C is T4 ligase sealing. Cycling drives the mixture toward the assembled product, which no longer has a BsaI site to cut.

NOTES

- **The ligase buffer supplies the ATP.** It is the component most often forgotten, and without it nothing ligates.
- 25 cycles is the default. Fewer (we often use 15) is faster but less efficient.
- Mix the DNAs equimolar if you can. If your preps are consistent, do not bother normalising.
- Miniprep, gel-purified and Zymo-cleaned DNA all work.
- Scaling up or down is fine as long as the buffer ends at **1×** and the DNA is not so concentrated that it inhibits the ligase.

Alternative. A straight 2 h incubation at 37 °C also works, and is popular.